

Fluoroscopic Evaluation of Duodenal Diseases

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Abbreviations: D1 = duodenal bulb, D2 = the descending duodenum, D3 = the transverse duodenum, D4 = the ascending duodenum, GI = gastrointestinal, NSAID = nonsteroidal anti-inflammatory drug

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SA-CME LEARNING OBJECTIVES

After completing this journal-based SA-CME activity, participants will be able to:

- Discuss the strengths, limitations, and techniques of fluoroscopy in evaluation of duodenal abnormalities.
- Describe the normal anatomy and common congenital variants of the duodenum.
- List the fluoroscopic imaging appearance of common infectious, inflammatory, and neoplastic conditions involving the duodenum.

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The duodenum can be affected by a variety of abnormalities because of its development during embryogenesis and its dual intra- and retroperitoneal location. If small bowel embryogenesis is disturbed, congenital errors occur. Although some congenital variants may be asymptomatic and inconsequential to the patient, other anomalies can result in life-threatening emergencies such as malrotation, leading to midgut volvulus. Many infectious processes affect the duodenum, including duodenal ulcers and opportunistic infection in patients with HIV/AIDS or Crohn disease. Small bowel malignancies are uncommon but important to recognize, because the duodenum can be involved in polyposis syndromes or the development of primary adenocarcinoma, neuroendocrine tumors, lymphoma, and metastasis. Although endoscopy is currently the most used diagnostic method to assess the lumen of the upper gastrointestinal tract, fluoroscopy is a valuable adjunct technique and the study of choice for many diseases, specifically those for which anatomic and functional information is required. Fluoroscopy is also commonly used postoperatively to assess for complications such as obstruction and extraluminal leaks. Compared with endoscopy, fluoroscopy is an inexpensive and noninvasive technique that provides salient anatomic information and allows delineation of the duodenal mucosa and assessment of real-time duodenal motility. The authors examine the broad spectrum of conditions that can involve the duodenum, including congenital, infectious, inflammatory, and neoplastic abnormalities, and review their typical appearances at fluoroscopy.

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Introduction

The duodenum is uniquely positioned in the intraperitoneal and retroperitoneal abdomen and is close to multiple other intra-abdominal structures. As a result, it can be affected by many primary and secondary diseases including traumatic, infectious, inflammatory, and neoplastic abnormalities (Table). Because development of the duodenum involves counterclockwise rotation of the midgut during embryogenesis, numerous congenital variants can occur if this process is interrupted. Although endoscopy, CT, and MR enterography are used for evaluation of the gastrointestinal (GI) tract, fluoroscopy is a noninvasive and inexpensive method to examine the duodenum. Thorough knowledge of optimal fluoroscopic techniques, normal anatomy, and

TEACHING POINTS

- A double-contrast technique is preferred in most scenarios. The patient first swallows effervescent granules with a small sip of water, then rapidly gulps barium to distend the esophagus. When the effervescent granules contact the stomach, they release gas and cause distention, which effaces the normal bowel folds and allows evaluation of gastric and bowel mucosa.
- One excellent sign of benignancy in both gastric and duodenal ulcers is a thin (<1 mm) radiolucent line at the neck of the ulcer that is visible during a barium examination of the ulcer in profile. The radiolucent line represents a thin rim of undermined bowel mucosa known as the Hampton line.
- Multiple postbulbar ulcers should raise suspicion for Zollinger-Ellison syndrome.
- Bowel obstruction due to superior mesenteric artery syndrome is relieved when the patient is placed in the prone or left lateral decubitus position.
- Bouveret syndrome is an uncommon variant of gallstone ileus, which occurs when a large gallstone erodes into the stomach or duodenum and causes obstruction of the outlet.

the typical appearance of various duodenal diseases at fluoroscopy is essential to make accurate diagnoses and provide quality patient care. In this article, we examine the fluoroscopic techniques that are recommended for an upper GI tract examination and describe the conventional anatomy of the duodenum, common anatomic variants, and the fluoroscopic imaging features of a variety of duodenal abnormalities.

Embryology and Anatomy

The GI tract develops from the foregut, midgut, or hindgut. The duodenum is derived from the foregut, which forms the esophagus, stomach, and duodenum up to the ampulla of Vater, and the midgut, which forms the distal duodenum through the proximal two-thirds of the transverse colon. During weeks 4 and 5 of development, the small and large intestines rapidly outgrow the available space in the abdominal cavity and eventually herniate outside the abdomen and into the umbilical cord. The herniated intestine rotates 90° counterclockwise around the mesentery, and the proximal bowel moves to the right while the distal bowel moves to the left. At 10 weeks of gestation, the bowel returns to the abdominal cavity and rotates an additional 180° counterclockwise (1,2).

The duodenum is the first part of the small intestine and connects the stomach with the second part of the small intestine, the jejunum. The duodenum begins at the duodenal bulb and ends at the ligament of Treitz, which is sometimes referred to as the duodenojejunal junction.

The duodenum is divided into four parts on the basis of anatomic orientation: the duodenal bulb (D1), descending duodenum (D2), transverse duodenum (D3), and ascending duodenum (D4) (Fig 1A). D1 is intraperitoneal for its first few centimeters before curving superiorly and posteriorly into the retroperitoneal abdomen (3). The remainder of the duodenum is retroperitoneal. The duodenojejunal junction is located to the left of the vertebral column, posterior to the stomach, and at the level of D1 (4).

At fluoroscopic examination, D1 has either smooth nearly featureless mucosa or a fine reticular mucosal pattern (Fig 1B) (5). D2–D4 have numerous parallel mucosal folds, also known as valvulae conniventes or plicae circulares. These mucosal folds are characteristic of the small bowel and extend from D2 through the jejunum before disappearing in the distal ileum (6).

Fluoroscopic Technique

A fluoroscopic upper GI series is an x-ray examination to evaluate the structure and function of the esophagus, stomach, and duodenum, for which the patient ingests barium or water-soluble contrast material. Barium, which is highly attenuating, is used unless there is concern for perforation, in which case, a water-soluble contrast agent is used. The upper GI fluoroscopic series can be performed with a “single-contrast” technique (contrast material alone) or a “double-contrast” technique (contrast material and effervescent granules). A single-contrast technique may be preferable in certain clinical situations such as in patients with limited mobility or a large body habitus and in those who have undergone restrictive gastric surgery. If the purpose of the study is assessment for perforation, obstruction, or dysmotility, a single-contrast technique may be sufficient.

A double-contrast technique is preferred in most scenarios. The patient first swallows effervescent granules with a small sip of water, then rapidly gulps barium to distend the esophagus. When the effervescent granules contact the stomach, they release gas and cause distention, which effaces the normal bowel folds and allows evaluation of gastric and bowel mucosa. Fluoroscopic evaluation of the esophagus and stomach is performed with a technique described by Levine et al (5,7), with the patient in a supine left posterior oblique position. Spot air contrast images of the duodenum are acquired with the patient in the left posterior oblique and left lateral positions. The patient then moves into the prone right anterior oblique position to acquire spot images of D1 (a compression paddle can be used in this position.), and a sweep through the

Clinical and Imaging Features of Duodenal Diseases

Condition	Clinical Features	Imaging Features
Malrotation	Asymptomatic Bilious vomiting in an infant	Abnormal location of the ligament of Treitz Duodenum and duodenojejunal junction to right of vertebral column Corkscrew appearance of duodenum
Duodenum inversum	Symptoms of obstruction	D3 turns right and courses superiorly before crossing midline above the pancreas Duodenojejunal junction normally located (unlike in malrotation)
Diverticula	Diverticulitis Symptoms of obstruction may occur with intraluminal diverticulum	Extraluminal collection of barium contiguous with the lumen Barium-filled saclike projection originating in the duodenal lumen
Familial visceral myopathy	Symptoms of obstruction	Megaduodenum
Annular pancreas	Recurrent pancreatitis Gastric outlet obstruction	Annular extrinsic compression D2
Duplication cyst	Asymptomatic Infection GI bleeding	Extramucosal mass
Ulcer	Burning or postprandial pain Peritonitis with rupture	Hampton line Cloverleaf bulb Extraluminal contrast material with perforation
Duodenitis	Pain and nausea	Thickening of the mucosal folds Nodularity Erosions
Crohn disease	Pain Symptoms of obstruction	Cobblestoning Luminal stenosis Fissures
Cytomegalovirus	Pain Nausea and vomiting GI bleeding	Thickening of the mucosal folds Ulcerations Erosions
Kaposi sarcoma	Symptoms of obstruction Malabsorption	Polypoid submucosal mass Irregular thickening
Aphthous erosion	Abdominal pain	Small oval lucency with central dot of contrast material
Adenocarcinoma	Pain, weight loss, gastrointestinal (GI) bleeding	Luminal narrowing in the short segment, with shouldering and ulceration
GI stromal tumor	Asymptomatic, GI bleeding, abdominal pain	Smooth submucosal filling defect
Peutz-Jeghers syndrome	Asymptomatic to symptoms of obstruction	Pedunculated, lobulated, or sessile polyps
Familial adenomatous polyposis	Asymptomatic to GI hemorrhage	Innumerable polyps
Zollinger-Ellison syndrome	Pain, weight loss, diarrhea, nausea, vomiting	Multiple benign-appearing ulcerations

ligament of Treitz is performed. If necessary, the patient may move into the right lateral decubitus position to allow contrast material to pool into D1. At the same time, the fluoroscopist can evaluate the duodenum for peristaltic abnormalities or mucosal disease. If air contrast images of the duodenum were not optimal in the previous positions, the patient can be returned to the left

posterior oblique or left lateral position at the end of the upper GI examination, in which additional spot images of the distended duodenum may be obtained (8). Although these views are standard in routine duodenal evaluation, the examination for each patient may be tailored on the basis of the patient's mobility or anatomy or to answer a specific clinical question.

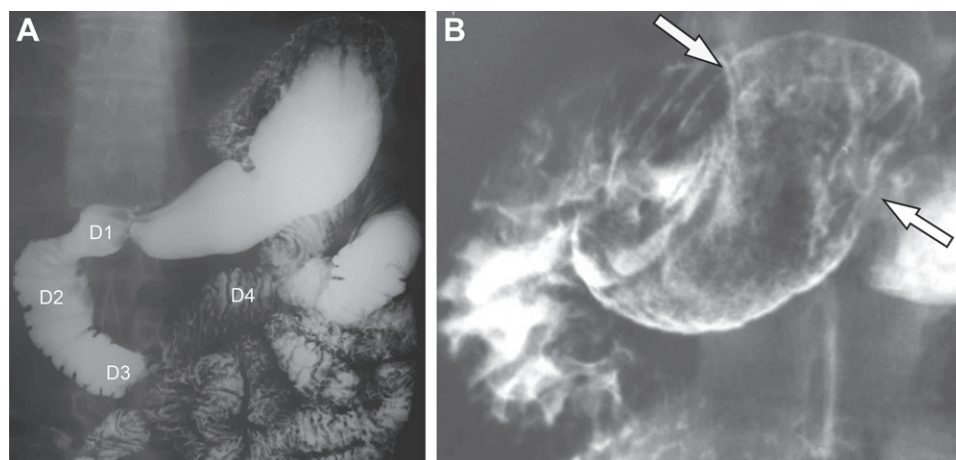


Figure 1. Normal appearance of duodenum in a 35-year-old man. (A) Anteroposterior fluoroscopic image of the upper GI tract shows the duodenum divided into four sections including the duodenal bulb (D1) and the descending (D2), transverse (D3), and ascending (D4) portions and ending at the ligament of Treitz. (B) Anteroposterior double-contrast fluoroscopic image shows the normal appearance of D1 (arrows) as a nearly featureless, fine reticular pattern. The remainder of the duodenum has multiple parallel folds and measures approximately 25 cm.

Congenital Variants

Intestinal Malrotation

Intestinal malrotation is a congenital anomaly that results from abnormal rotation of the midgut as it returns to the abdominal cavity during embryogenesis. Any deviation from the normal 270° counterclockwise rotation of the midgut is considered malrotation, although the degree of deviation is on a spectrum and can be complete or incomplete (9). The ligament of Treitz is normally positioned directly anterior to or to the left of the vertebral column at the level of the D1 apex. In malrotation, the ligament of Treitz is abnormally positioned (Fig 2). In infants with bilious emesis, the most concerning diagnostic consideration is midgut volvulus, which is a life-threatening condition that is associated with malrotation and can result in bowel necrosis if it is not treated with urgent surgical repair (10). At fluoroscopy, midgut volvulus appears as proximal bowel dilatation to the level of D2 or D3, with an associated spiral or corkscrew appearance of bowel loops just beyond the dilated bowel. In a patient with bilious vomiting, malrotation without this imaging appearance may represent intermittent volvulus.

Duodenum Inversum

Duodenum inversum is a rare congenital anomaly in which the normal curvature of the duodenum is reversed. Instead of D3 turning left and crossing the midline, it turns right and courses superiorly before curving to the left and crossing the midline above the pancreas (Figs 3, E1) (11). Unlike its position in patients with malro-

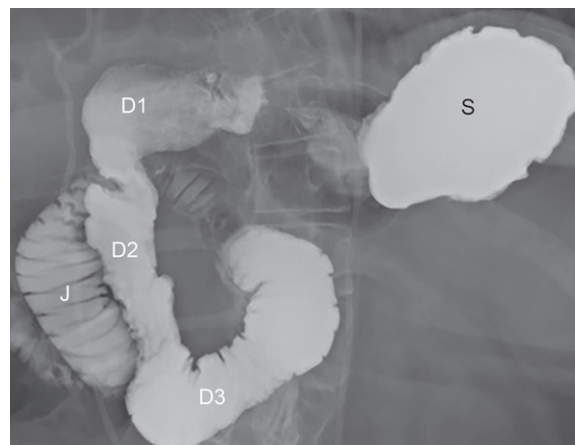


Figure 2. Malrotation in a 21-year-old woman with chronic abdominal pain. Anteroposterior fluoroscopic image of the upper GI tract shows the abnormal course of D4 (midline location) and the jejunum (J), which is located in the right upper quadrant. The duodenojejunal junction is abnormally located to the right of the midline. S = stomach.

tation, the duodenojejunal junction is normally positioned. Duodenum inversum is usually asymptomatic, although some patients may present with obstructive symptoms such as epigastric pain, nausea and vomiting, and abdominal distention. Because of the stasis of ingested contents in the duodenum, patients may be at increased risk of peptic ulcers and pancreatitis (12). Duodenum inversum is not associated with midgut volvulus.

Duodenal Diverticula

Diverticula occur when mucosal and submucosal layers herniate through a muscular defect in the bowel wall (13). After the colon, the duodenum

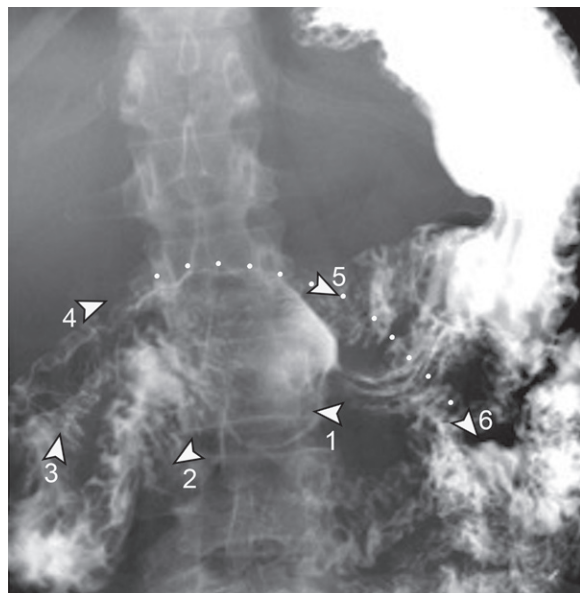


Figure 3. Duodenum inversum in a 24-year-old woman who presented with abdominal pain. Anteroposterior fluoroscopic image of the upper GI tract shows the distal duodenum to the right of D2 so that the duodenojejunal junction assumes a high medial position in the left retroperitoneum. The course of the duodenum inversum (arrowheads 1–3) and the posterior course (arrowheads 4–6, dotted line) are shown.

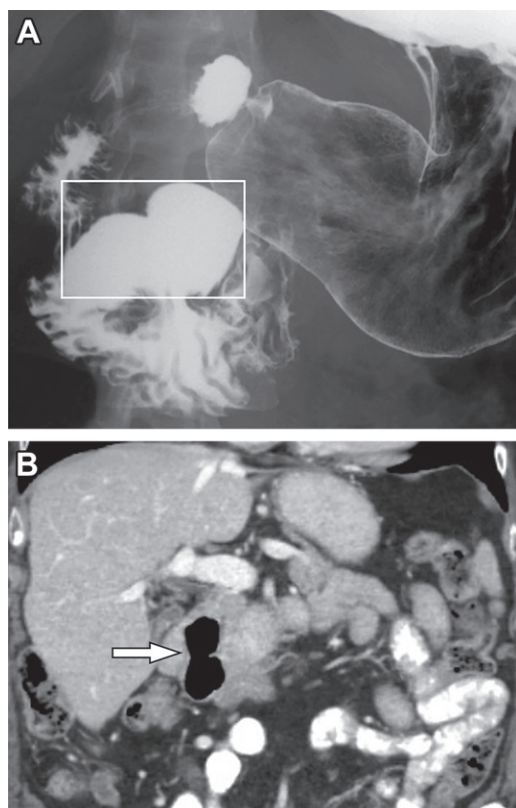


Figure 4. Duodenal diverticulum. (A) Anteroposterior fluoroscopic image of the upper GI tract in a 66-year-old woman who presented with melena shows a contrast material-filled outpouching (box), arising from the mesenteric side of the D3 duodenal lumen. (B) Coronal CT image (soft-tissue window) shows a diverticulum as an air-filled outpouching (arrow), arising from D2.

is the most common location for diverticula, which usually occur on the mesenteric side along the medial wall of D2 or the superior wall of D3. Ninety percent of cases of diverticula are asymptomatic, but rare complications include diverticulitis or incompetence of the biliary sphincter if the diverticulum is located near the papilla of Vater (14). At fluoroscopy, duodenal diverticula appear as extraluminal collections of barium that are contiguous with the duodenal lumen and can change configuration throughout the examination (Fig 4A). At CT, they appear as saccular outpouchings from the duodenum and may contain air, fluid, or food material (Fig 4B) (15). Although it is rare, the duodenum may develop numerous diverticula with an imaging appearance that is comparable to that of colonic diverticulosis (Fig E2).

Intraluminal Diverticula

An *intraluminal diverticulum* is a sac-like projection typically in D2 that develops from gradually increasing protrusion of a congenital mucosal diaphragm or web (16). As the diverticulum enlarges because of repetitive peristalsis, patients can develop symptoms of nausea, vomiting, and postprandial pain. The classic appearance on fluoroscopic images of the upper GI is the “windsock” sign, which is a contrast material-filled sac in the duodenal lumen that is surrounded by a narrow lucent line that represents the mucosa of the diaphragm or the web (Figs 5, E3). The fundus of the diverticulum can extend as far as D4 (17).

Visceral Myopathy

Visceral myopathy is a rare neuromuscular condition that is characterized by muscle cell degeneration and fibrosis, leading to impaired intestinal function and motility. This can result in chronic dilatation and pseudo-obstruction of the GI tract. Clinical manifestations of visceral myopathy range from asymptomatic to repeated episodes of bowel obstruction (18). Evaluation of affected patients with barium contrast material demonstrates marked duodenal dilatation (megaduodenum), with decreased motility and contrast material stasis (Fig 6) (19).

Annular Pancreas

An *annular pancreas* is characterized by a band of pancreatic tissue completely or incompletely encircling D2 because of incomplete rotation of the ventral bud of the pancreas during embryogenesis (20). Approximately 50% of cases manifest during infancy, with symptoms of gastric outlet obstruction. The remaining 50% are diagnosed in adults either incidentally or in those

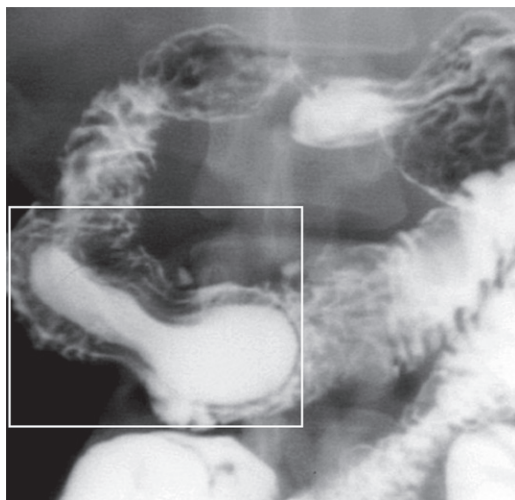


Figure 5. Intraluminal diverticulum in a 40-year-old woman with intermittent nausea and vomiting. Fluoroscopic image shows a barium-filled structure (box) that is outlined by a thin radiolucent line in the early D3.

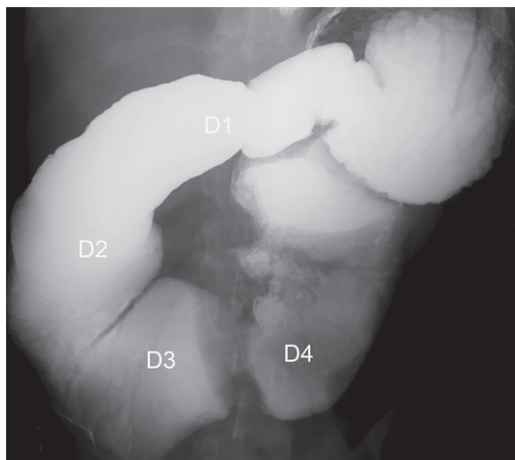


Figure 6. Familial visceral myopathy or megaduodenum in a 25-year-old man with recurrent vomiting. Fluoroscopic image shows marked dilatation of the duodenum (D1–D4). The apparent narrowing at the D3–D4 junction is caused by a coursing vessel.

with symptoms of partial gastric outlet obstruction, upper GI bleeding from peptic ulcer disease, or recurrent pancreatitis (3). Fluoroscopic evaluation shows a smooth extrinsic compression of D2, and at CT, an enlarged pancreatic head is seen encircling D2 (Fig 7) (20).

Duplication Cyst

Enteric duplication cysts result from incomplete recanalization of the GI tract during embryogenesis. The duodenum is the least-common location, accounting for approximately 5% of all cases (21). Duodenal duplication cysts are usually located posteromedially to D2 and D3 and can cause nausea, vomiting, and abdominal pain. In rare cases, they can be complicated by

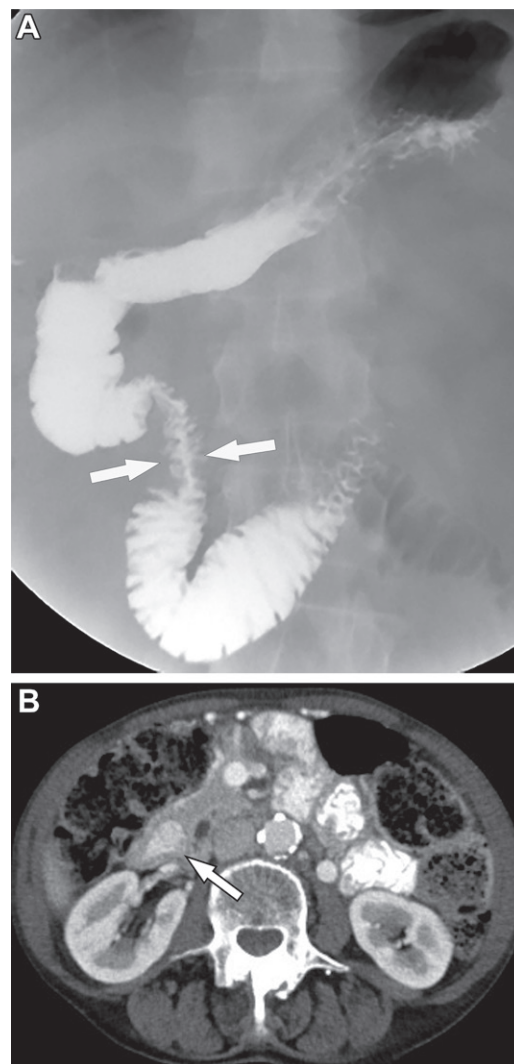


Figure 7. Annular pancreas in a 55-year-old man with abdominal distention and pain. (A) Anteroposterior fluoroscopic image of the upper GI tract shows focal narrowing of D2 (arrows), with upstream dilatation. The mucosal folds of the duodenum are preserved, which helps in differentiation of this benign entity from a neoplastic process. (B) Axial CT image (soft-tissue window) shows the contrast-material-filled duodenum (arrow) encircled by pancreatic tissue.

inflammation, infection, bleeding, or intussusception (22). Barium evaluation may show a polypoid submucosal mass or a smooth extrinsic compression of the duodenum (Fig 8). If the cyst communicates with the GI tract, there may be a tubular barium-filled mass adjacent to the duodenal lumen that resembles a duodenal diverticulum (23).

Brunner Gland Hyperplasia

Brunner gland hyperplasia results from the overgrowth of submucosal alkaline-secreting Brunner glands that appear as one or more small nodules in the proximal duodenum at endoscopy. At fluoroscopy, Brunner gland hyperplasia

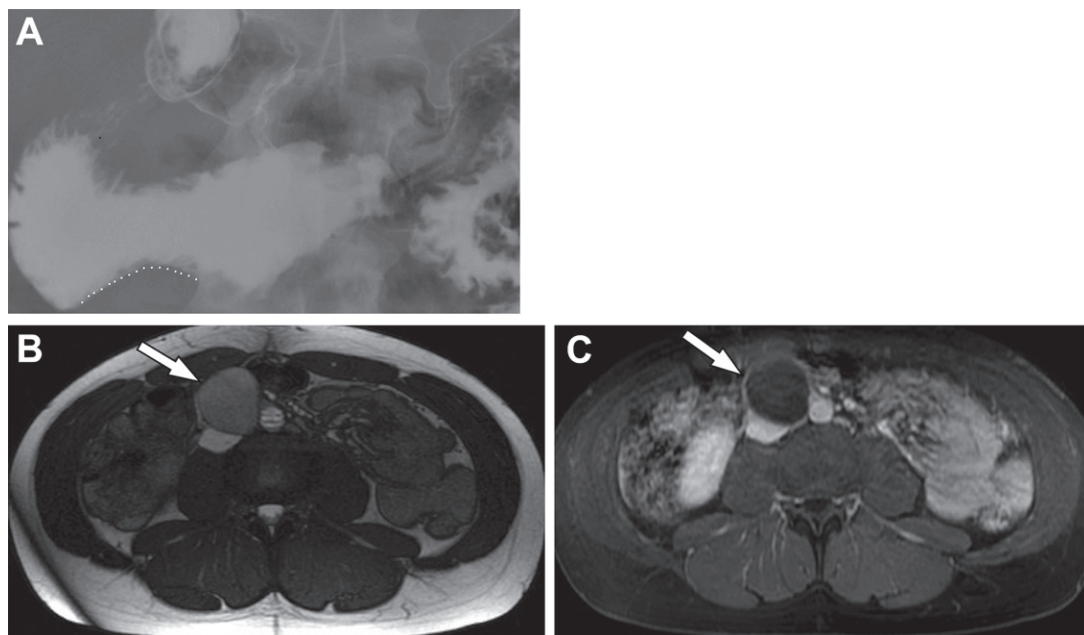


Figure 8. Infected duplication cyst in a 27-year-old man who presented with abdominal pain. (A) Fluoroscopic image shows the inferior indentation (dotted line) in D3, with obtuse margins and preserved mucosal folds, which are characteristic of extrinsic compression. (B, C) Axial T2-weighted fast spin-echo fat-saturated (B) and axial T1-weighted fast spin-echo fat-saturated contrast-enhanced (C) MR images show a round moderately hyperintense (arrow in B) and hypointense (arrow in C) lesion with peripheral enhancement that is contiguous with the inferior wall of D3.

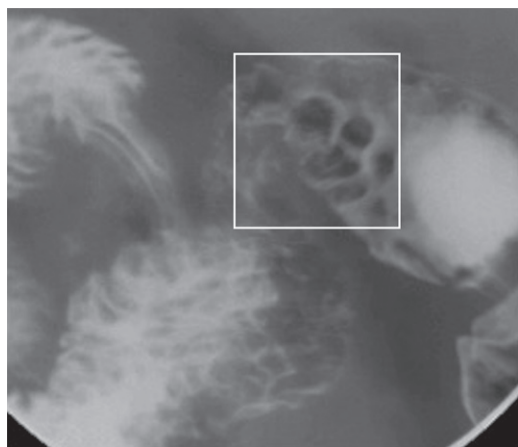


Figure 9. Brunner gland hyperplasia in a 57-year-old man with right upper quadrant pain. Coned-down fluoroscopic image of the upper GI tract shows multiple well-rounded filling defects in D1 (box).

manifests as multiple filling defects that project into the lumen (Fig 9). The differential diagnosis for multiple small nodules in the duodenum includes familial adenomatous polyposis, Peutz-Jegher syndrome, nodular lymphoid hyperplasia, nodular duodenitis, lymphoma, and metastatic disease. Patients are usually asymptomatic but can develop bleeding, obstruction, and pancreatitis. A hyperplastic Brunner gland with a diameter greater than 5 mm is referred to as a Brunner gland hamartoma, which can be complicated by hemorrhage or obstruction (24).

Inflammatory and Infectious Conditions

Aphthoid Lesions

Aphthoid lesions are submucosal ulcers on the surface of hyperplastic intestinal lymphoid follicles. They result from a variety of conditions including infectious enterocolitis, Crohn disease, polyarteritis nodosa, and Behcet disease. At fluoroscopy, the aphthae appear as small oval areas of lucency, with a central collection of contrast material that represents focal erosion surrounded by a halo of inflamed and edematous mucosa (Fig 10) (25). Because of the superficial mucosal character of these erosions, the double-contrast (barium and air) technique is needed to optimize luminal distention and mucosal coating.

Nodular Lymphoid Hyperplasia

Nodular lymphoid hyperplasia is a rare and benign type of lymphoproliferative disease that forms from lymphoid aggregates in the bowel mucosa. It is most frequently found in the terminal ileum, which contains the highest concentration of lymphoid follicles. In pediatric patients, it is associated with viral infections or food allergies. In adults, it has been associated with common variable immune deficiency and *Giardia* or *Helicobacter pylori* infection (26,27). Lymphoid hyperplasia can also be a normal variant or incidental finding, although some patients experience weight loss, diarrhea,



Figure 10. Aphthae in a 45-year-old man with dyspepsia. Fluoroscopic image shows multiple small contrast material-filled irregularities in the D1 mucosa (arrowheads).

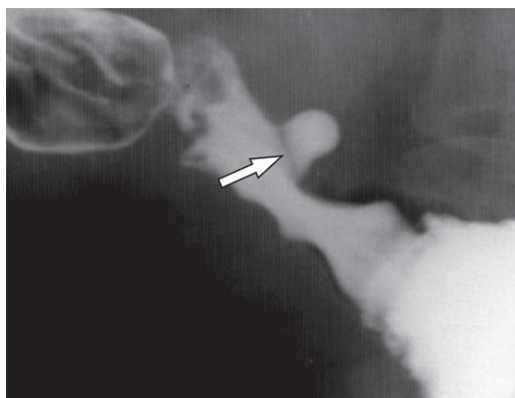


Figure 12. Duodenal ulcer in a 51-year-old man with abdominal pain. Coned-down fluoroscopic image shows an ulcer in D1, with a well-defined thin mucosal rim known as the Hampton line (arrow) that is overhanging the ulcer's crater.

and epigastric pain. At fluoroscopy, nodular lymphoid hyperplasia appears as numerous small (<3 mm) mucosal filling defects (Fig 11).

Duodenal Ulcer

Duodenal ulcers are two to three times more common than gastric ulcers, but unlike gastric ulcers, they are almost always benign. Common associations include *H. pylori* infection and use of nonsteroidal anti-inflammatory drugs (NSAIDs), alcohol, steroids, and tobacco. Approximately 95% of duodenal ulcers form in D1. They appear as round or ovoid collections of contrast material projecting beyond the expected bowel wall, with radiating folds converging at the edge of these collections (Fig 12) (28).

One excellent sign of benignancy in both gastric and duodenal ulcers is a thin (<1 mm) radiolucent line at the neck of the ulcer that is



Figure 11. Nodular lymphoid hyperplasia in a 75-year-old woman who presented with epigastric pain. Fluoroscopic image shows multiple nodular defects in D1 (box) that are consistent with nodular lymphoid hyperplasia.

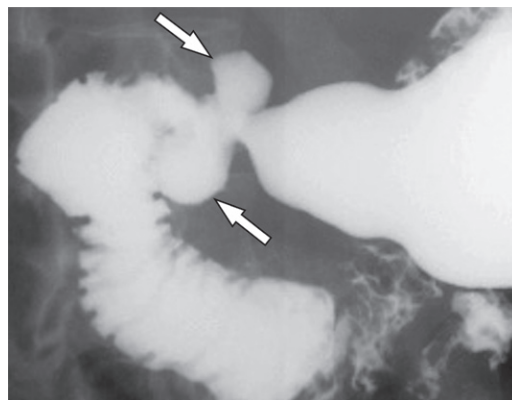


Figure 13. Cloverleaf configuration of D1 in a 53-year-old woman with chronic anemia, esophageal discomfort, and severe reflux. Fluoroscopic image from an upper GI series shows the cloverleaf configuration of D1 (arrows), which occurs secondary to chronic inflammation and scarring.

visible during a barium examination of the ulcer in profile. The radiolucent line represents a thin rim of undermined bowel mucosa known as the Hampton line (Fig E4) (29). Although it is reliable, this sign is not sensitive for diagnosis of ulcers, and its absence does not indicate malignancy. Another radiographic sign that is associated with duodenal ulcers is the “cloverleaf” deformity, which forms after repeated episodes of ulceration have led to scarring and fibrosis of D1 (Fig 13). Areas of normal duodenal tissue bulge between areas of fibrosis, resulting in pseudodiverticula and the characteristic cloverleaf appearance at fluoroscopy (30).

Zollinger-Ellison Syndrome

Zollinger-Ellison syndrome is a severe peptic ulcer disease that results from gastric acid hypersecretion from a gastrin-secreting tumor that is usually located in the pancreas or duodenum. Most

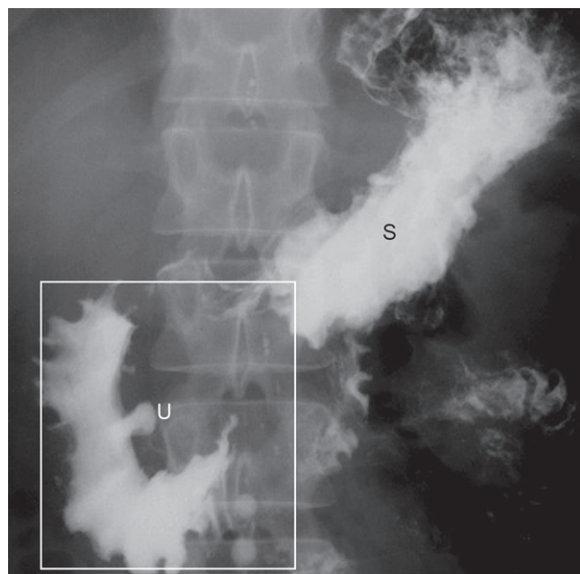


Figure 14. Zollinger-Ellison syndrome in a 27-year-old woman with a long history of dyspepsia and peptic ulcer disease and a family history of bleeding ulcer diathesis. Anteroposterior fluoroscopic image of the upper GI tract shows thickening of the duodenal folds (box), with loss of the normal mucosal pattern and a postbulbar ulcer (U). Diffuse thickening of the gastric fold is also present. S = stomach.

gastrinomas form in the *gastrinoma triangle*, an anatomic area in the abdomen that is formed superiorly by the confluence of the cystic and common bile ducts, inferiorly by D2 and D3, and medially by the junction of the pancreatic neck and body. Twenty to thirty percent of gastrinomas occur in association with multiple endocrine neoplasia type 1, and the rest occur sporadically (31). More than 90% of patients with Zollinger-Ellison syndrome develop peptic ulcers, primarily in D1. Multiple postbulbar ulcers should raise suspicion for Zollinger-Ellison syndrome (Fig 14) (32).

Perforated Ulcer

Duodenal ulcers can be asymptomatic or result in burning or postprandial pain that is relieved by eating. Complications include hemorrhage, perforation, obstruction, or formation of a fistula. If there is strong clinical suspicion for perforation of a duodenal ulcer, patients typically undergo CT to assess for pneumoperitoneum, extraluminal leakage of water-soluble oral contrast material, or fluid collections. An upper GI series with water-soluble contrast material may be performed to evaluate the location and extent of the perforation and to determine if the perforation has sealed. The presence of extraluminal contrast material during the upper GI examination allows confirmation of bowel perforation and identification of the site of the perforation (Fig 15).

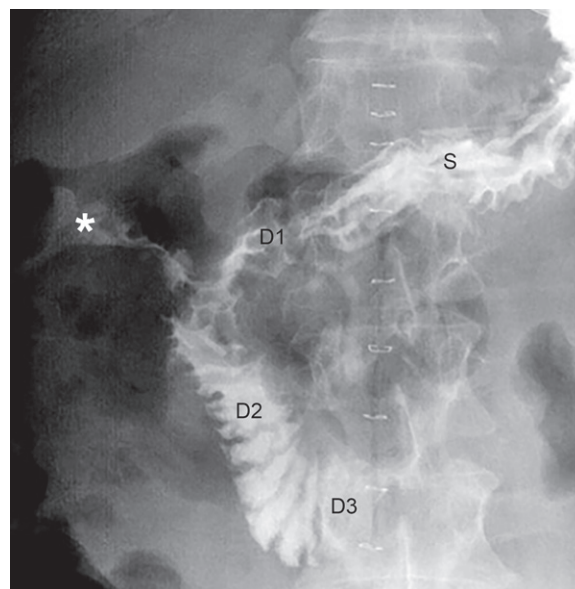


Figure 15. Perforated duodenal ulcer in a 62-year-old man who presented with severe abdominal pain. Fluoroscopic image from an upper GI series shows an extraluminal leak of oral contrast material (*) from D1. The contrast material opacifies the stomach (S) and the remainder of the duodenum (D1–D3).

Duodenitis

Duodenitis is inflammation of the duodenum, most commonly caused by infection with *H. pylori* or the use of NSAIDs. Other causes include pancreatitis, Crohn disease, radiation, viral or bacterial infections, atypical infections including fungal infection and mycobacterium avium-intracellulare infection, and granulomatous disease. The fluoroscopic appearance of duodenitis is generally similar for all causes and includes thickening of the mucosal folds, erosions, and bulb deformity (Figs 16, E5). Thickening of the mucosal folds is the most common imaging finding in patients with duodenitis but is also less specific than erosions for diagnosis of duodenitis, because fold thickening can also occur in patients with hemorrhage and edema. Very small erosions and mucosal nodularity are best shown with the double-contrast technique (33).

Crohn Disease

Crohn disease is an idiopathic chronic inflammatory disease that is characterized by discontinuous transmural inflammation of the GI tract that occurs anywhere from the mouth to the anus. The terminal ileum and proximal colon are most frequently involved, and the duodenum is only affected in 0.5%–4% of cases (34). Most patients with duodenal Crohn disease also have concurrent involvement of the terminal ileum or colon. The most common pattern of involvement is continuous inflammation of the gastric antrum, the

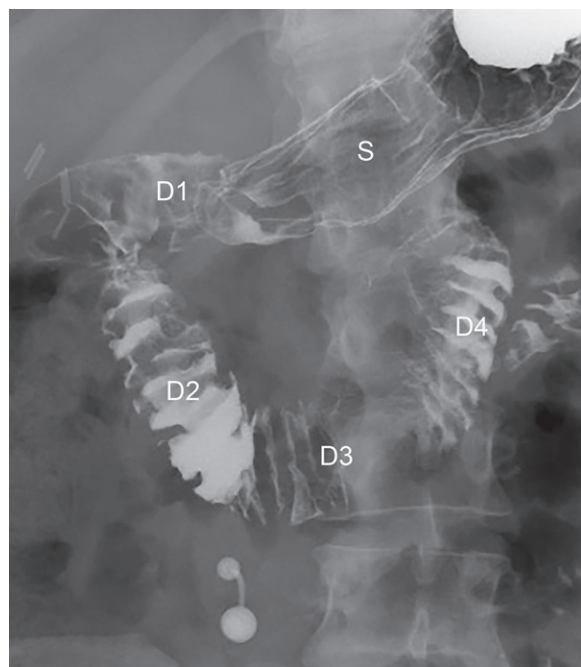


Figure 16. Endoscopically proven duodenitis in a 44-year-old woman who presented with abdominal pain, nausea, and vomiting. Fluoroscopic image shows smooth segmental mucosal thickening involving D2–D4. S = stomach.

pylorus, and the proximal duodenum, with short segmental strictures that are surrounded by ulcerated or nodular mucosa (35). Repeated inflammation of the gastric antrum and the duodenal bulb leads to a tubular deformed shape of the stomach as it progressively tapers from the antrum to the pylorus, which is referred to as the “ram’s horn” or “pseudo-Billroth I” sign. Repeated inflammation can also eventually lead to severe luminal stenosis, resulting in the “string sign,” which refers to the string-like appearance of the contrast material-filled bowel due to severe narrowing. Other imaging changes include mural thickening, cobblestoning, and aphthous lesions (Fig 17) (36).

HIV Infection

HIV infection of the duodenum appears at fluoroscopy as ulceration of the small bowel, villous atrophy, and tiny erosions (37). Opportunistic infections involving the duodenum in patients with HIV/AIDS include cytomegalovirus, cryptosporidiosis, tuberculosis, fungal infections, lymphoma, and Kaposi sarcoma (Fig 18). The incidence of disease increases as the CD4 cell count decreases. For example, Kaposi sarcoma occurs with a CD4 count lower than 350, tuberculosis and lymphoma occur at a count lower than 200, and cytomegalovirus colitis occurs at a count lower than 100 (38).

Cytomegalovirus Infection

Cytomegalovirus is an opportunistic viral infection that remains latent in immunocompetent patients

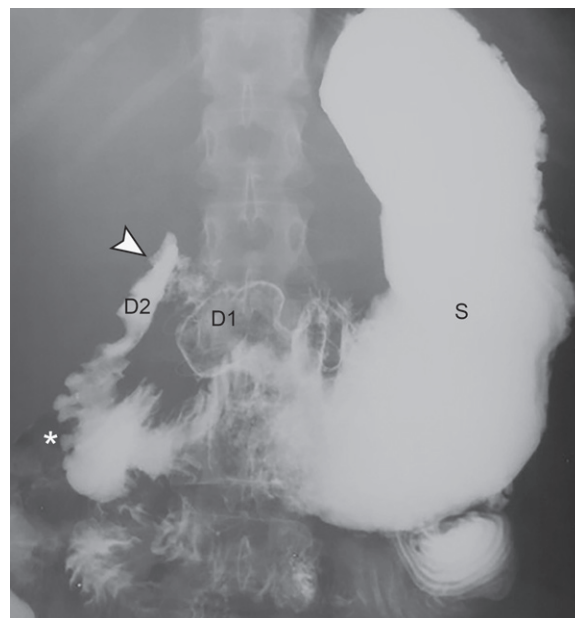


Figure 17. Crohn disease of the proximal duodenum in a 27-year-old woman with a history of multiple ileal resections. Fluoroscopic image of the upper GI tract shows marked fold thickening and irregularity in D2 and D3, resulting in luminal narrowing. Also visible is a “rose thorn ulcer,” which is a deep penetrating ulcer with fissuring (arrowhead), and mucosal ulceration (*), called a “collar button ulcer,” which is associated with ulcerating inflammatory disease. S = stomach.

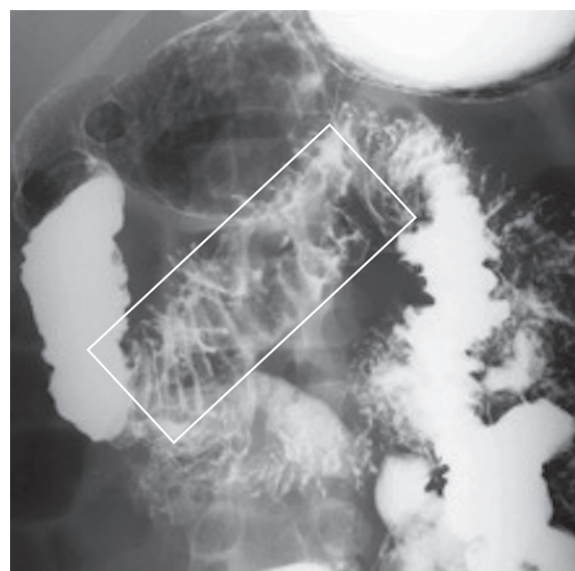


Figure 18. Cryptosporidiosis in a 27-year-old man with AIDS and epigastric pain. Anteroposterior fluoroscopic image from an upper GI series shows irregular mucosal thickening and irregularity of the duodenum (box) and jejunum.

but can become reactivated with immunosuppression. In patients with a CD4 count lower than 100, cytomegalovirus commonly affects the GI tract and usually manifests as cytomegalovirus colitis. Cytomegalovirus infection of the upper GI tract is less common but can cause esophagitis, gastritis, or duodenitis. Patients present

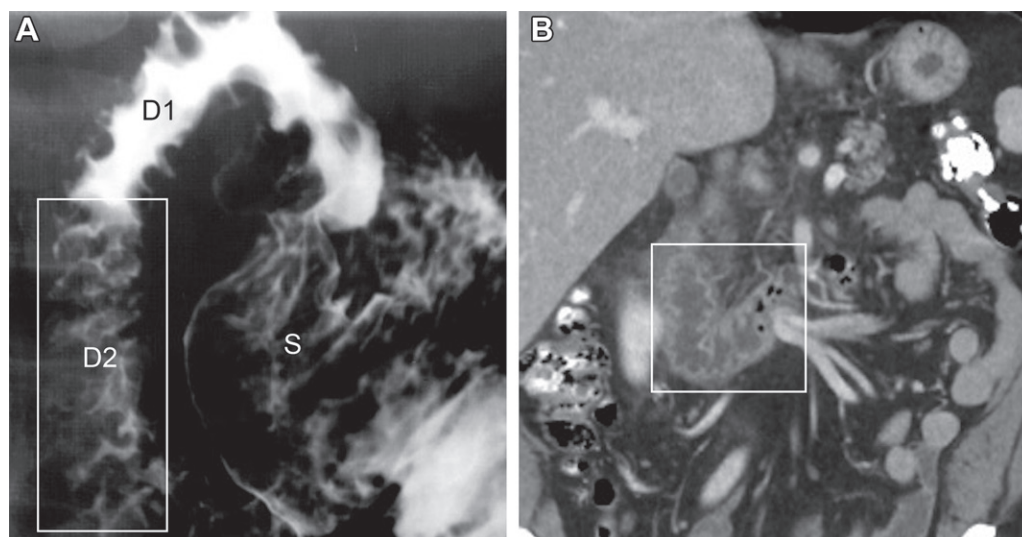


Figure 19. Cytomegalovirus duodenitis in a 43-year-old man who is HIV positive, with dyspepsia, nausea, and vomiting. **(A)** Fluoroscopic image shows thickened folds and small aphthous lesions involving D1 and D2 (box). S = stomach. **(B)** Coronal CT image (soft-tissue window) shows thickening of the duodenal wall (box), with surrounding inflammatory changes.

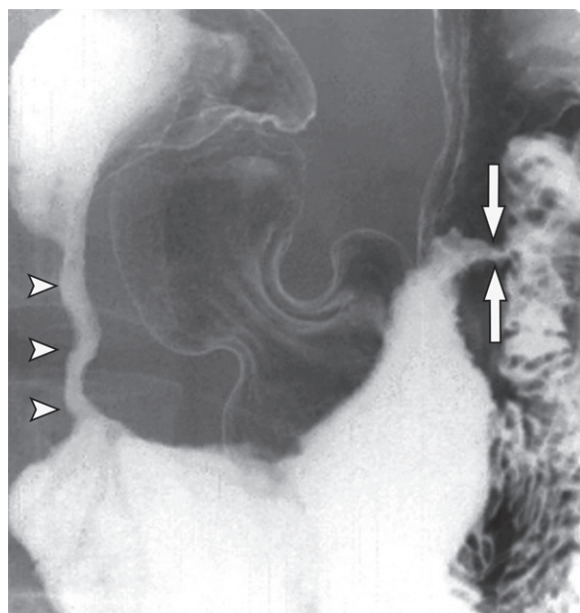


Figure 20. Miliary tuberculosis in a 45-year-old woman who presented with abdominal pain and early satiety. Coned-down fluoroscopic image from an upper GI series shows a long segment of severe narrowing of the proximal duodenum (arrowheads) and a short stricture of the distal duodenum (arrows).

with abdominal pain, GI bleeding, or nausea and vomiting. The fluoroscopic appearance of cytomegalovirus duodenitis is nonspecific and includes mucosal inflammation and thickening, ulcerations, and erosions (Fig 19). Rare complications include hemorrhage and bowel perforation (39,40).

Tuberculosis Stricture

Although the ileocecal region is the most common site for GI tuberculosis, the duodenum is a rare site for disease and usually results from

secondary pulmonary spread (41). The imaging features are related to the degree of inflammation, with mild cases showing symmetric mural thickening and severe cases demonstrating stricturing, nodularity, and ulceration (Fig 20). Upper GI studies may exhibit nonspecific inflammatory features such as mucosal ulcerations, luminal narrowing, proximal dilation, and extrinsic compression (42).

Neoplasms

Lipomas

GI lipomas are benign tumors composed of mature adipocytes, which can be found in the small and large bowel. Although GI lipomas are usually asymptomatic and only diagnosed incidentally, a pedunculated lipoma can act as a lead point for intussusception or obstruction (43). At fluoroscopy, lipomas appear as smoothly margined and compressible masses that are usually found in D3 (Fig 21) (44).

Adenomas and Villous Adenomas

Adenomas are benign precursors to adenocarcinomas and can undergo an adenoma-to-carcinoma sequence that involves inactivation of tumor suppressor genes and activation of oncogenes. One type of advanced adenoma is the villous adenoma, which is larger and more likely to degenerate into malignancy when compared with an early adenoma. When they are numerous, superficial villous adenomas cover a large surface area and are known as “carpet lesions.” The duodenum is the most common extracolonic location. At fluoroscopy, adenomas

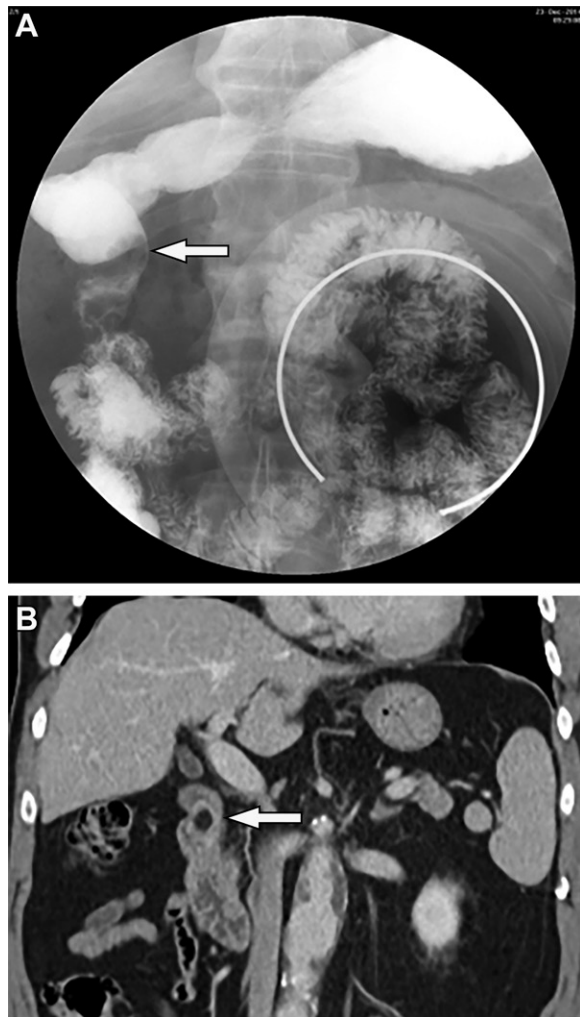


Figure 21. Duodenal lipoma in a 70-year-old woman who presented with vague abdominal discomfort. **(A)** Fluoroscopic image shows a smooth intraluminal filling defect (arrow) in D2. **(B)** Axial CT image (soft-tissue window) shows that the filling defect on the upper GI image corresponds to an intramural fat-attenuation mass (arrow), which is consistent with a lipoma.

appear as pedunculated or sessile rounded filling defects that form acute angles with the bowel wall (45). Villous lesions appear as lobulated or “soap bubble” lesions with a lacelike pattern because of barium insinuating into clefts between villous projections.

Familial Adenomatous Polyposis

Familial adenomatous polyposis is an autosomal-dominant syndrome caused by a germline mutation that deletes one copy of a tumor suppressor gene (*APC*) in the adenoma-to-carcinoma sequence, making affected patients much more likely to develop adenomas and adenocarcinomas. Patients with familial adenomatous polyposis develop innumerable GI adenomatous polyps, with duodenal adenomas occurring in up to 90% of patients (Fig 22) (46). Familial adenomatous polyposis carries a nearly 100% chance

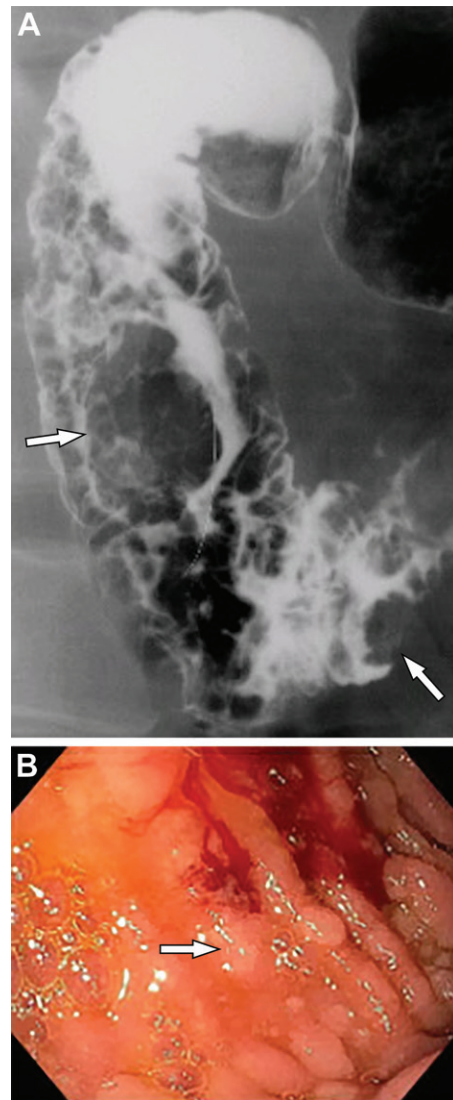


Figure 22. Familial adenomatous polyposis in a 40-year-old man who underwent prophylactic total colectomy at the age of 16 years and presented with early satiety and melena. **(A)** Coned-down fluoroscopic image shows multiple irregular polypoid filling defects of various sizes in D2–D4, representing numerous adenomatous polyps (arrows). **(B)** Upper endoscopic image shows numerous duodenal polyps (arrow), some of which are bleeding.

of degeneration to malignancy of colorectal polyps, although only 1%–5% of patients develop duodenal carcinoma (47).

Adenocarcinoma

Although adenocarcinomas of the small bowel are rare (<2% of all GI tract tumors), they account for approximately 42% of all malignant neoplasms of the small bowel (47). The overwhelming majority of duodenal adenocarcinomas are in the periampullary region of D2. As with other bowel carcinomas, adenocarcinoma of the duodenum appears at fluoroscopy as short-seg-



Figure 23. Peripapillary duodenal adenocarcinoma in the classic location in a 60-year-old man who presented with abdominal pain, vomiting, and weight loss. Fluoroscopic image shows the luminal narrowing caused by an annular mass (arrowheads), arising from the inferior aspect of the irregular papilla (*).

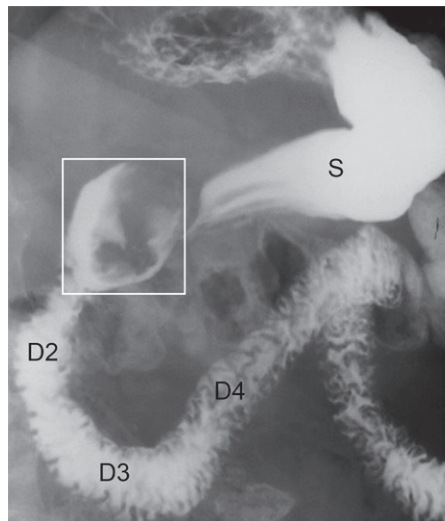


Figure 24. Adenocarcinoma involving D1 in a 76-year-old man with abdominal pain and melena. Fluoroscopic image from an upper GI series shows an irregular masslike filling defect in D1 (box), with normal contrast opacification of the distal duodenum and proximal jejunum. S = stomach.

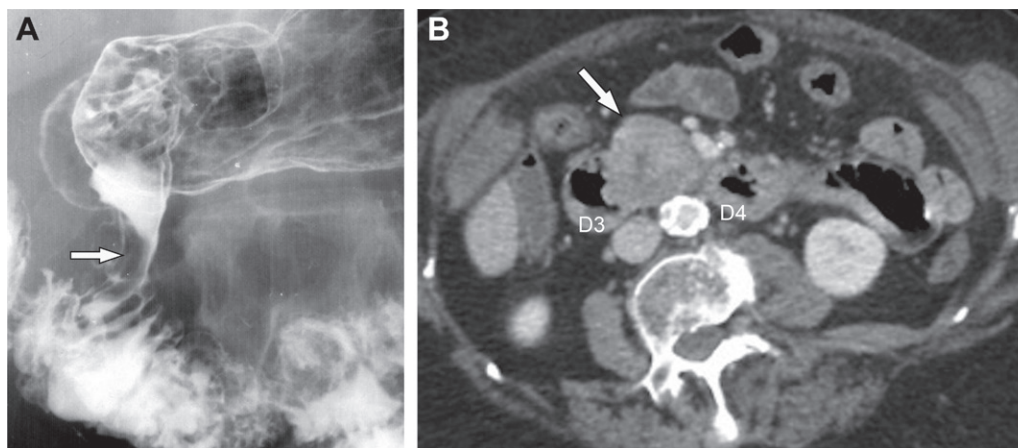


Figure 25. GI stromal tumor in the duodenum in a 59-year-old man who experienced 2 months of bloating and abdominal pain. (A) Fluoroscopic image from an upper GI series shows a well-circumscribed rounded extramural narrowing of the lateral wall of D2, with obtuse margins (arrow). (B) Axial CT image (soft-tissue window) shows an exophytic heterogeneously enhancing GI stromal tumor (arrow), arising from D2.

ment luminal narrowing with abrupt transition or as a polypoid and irregular mass (Figs 23, 24).

GI Stromal Tumors

GI stromal tumors are extramucosal intramural tumors that predominantly occur in the GI tract, yet they have been found occasionally in the mesentery, peritoneum, and omentum. Approximately 25% of all GI stromal tumors occur in the small bowel and are the most common small bowel mesenchymal tumors. Patients with GI stromal tumors most commonly present with abdominal pain, GI bleeding, or obstruction, although they can also be asymptomatic (48). At fluoroscopy, most GI stromal tumors appear as a smooth submucosal mass,

although ulcerated or necrotic GI stromal tumors may communicate with the bowel lumen (Fig 25).

Hamartomas

Peutz-Jeghers syndrome is an autosomal-dominant disorder that results in hamartomatous GI polyps and mucocutaneous macules (Fig 26). Individuals with Peutz-Jeghers syndrome are at increased risk for GI and extraintestinal malignancies, with a relative risk of 9.9–18 compared with that of the general population (49). The hamartomatous GI polyps most commonly occur in the small bowel, with a jejunal predominance, and become symptomatic between the 1st and 3rd decades of life (50). These polyps

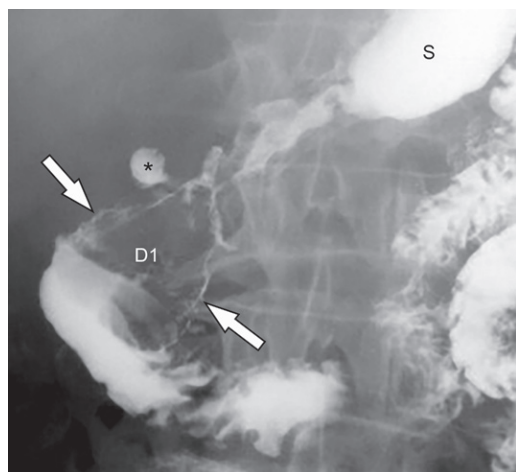


Figure 26. Brunner gland hamartoma in an 81-year-old man with a history of peptic ulcer disease and abdominal pain. Fluoroscopic image from an upper GI examination shows a large mucosal mass in D1 and proximal D2 (arrows) that is causing peripheral displacement of contrast material. Also visible is a collection of extraluminal contrast material in the posterior duodenal wall, with a well-defined thin mucosal rim traversing the neck, which is consistent with a benign ulcer (*). S = stomach.

can lead to bowel obstruction, intussusception, and anemia from bleeding. Peutz-Jeghers polyps have no distinct imaging appearance and can be pedunculated, lobulated, or sessile.

Lymphoma

Lymphoma is the most common malignancy of the small bowel and includes B-cell, T-cell, Burkitt, mucosa-associated lymphoid tissue, and Hodgkin lymphoma. At fluoroscopy, lymphoma can have many different manifestations, including diffuse or segmental ulceration or nodularity, aneurysmal dilatation, or intussusception. It can also appear as a single or multiple polypoid intraluminal masses. Mucosa-associated lymphoid tissue lymphoma is a low-grade non-Hodgkin lymphoma that typically arises in the stomach. Fewer than 30% of cases arise in the small intestine (51). Primary duodenal mucosa-associated lymphoid tissue lymphoma is rare and has similar imaging features to those of most other subtypes of lymphoma (Fig 27) (52). Of all lymphomas, T-cell lymphoma is most common in the small bowel and is more likely to have multifocal involvement and cause perforation (53).

Neuroendocrine Tumors

Neuroendocrine tumors of the duodenum are usually discovered incidentally during endoscopy. They usually occur sporadically and are nonfunctioning, although approximately one-third are functional and manifest with signs of Zollinger-Ellison syndrome including abdominal pain, nausea, vomiting, weight loss, and diarrhea. At

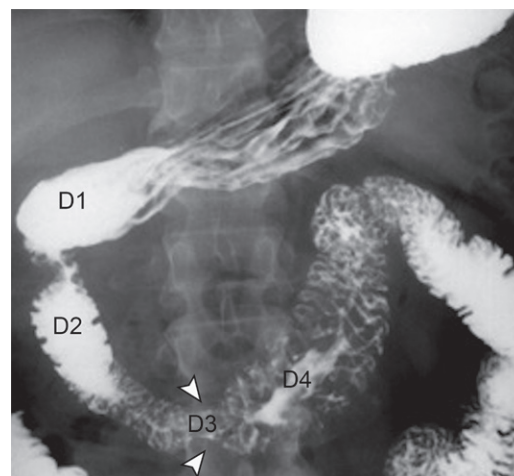


Figure 27. Mucosa-associated lymphoid tissue lymphoma in a 70-year-old man who presented with nausea and dyspepsia. Fluoroscopic image shows mucosal thickening, with ulcers in D3 and D4 (arrowheads).

fluoroscopy, they appear as a mural or polypoid mass projecting into the lumen, with overlying mucosal ulceration. In patients with multiple endocrine neoplasia type 1 and Zollinger-Ellison syndrome, there is diffuse thickening of the duodenal fold and multiple erosions and ulcers due to gastric acid hypersecretion (54),

Metastasis

Metastasis to the duodenum is rare and typically occurs in patients with a known primary malignancy and widespread metastatic disease (Fig 28). The imaging features of metastasis at fluoroscopy are nonspecific, overlap with those of other previously described malignant neoplasms, and include an exophytic or intramural mass, central necrosis, or focal ulceration (55). Metastasis to the duodenum usually occurs by means of direct extension and intraperitoneal seeding from pancreatic, gastric, colonic, and gynecologic malignancies, with far fewer instances of hematogenous spread from primary renal cell carcinoma, melanoma, breast cancer, and lung cancer (56).

Kaposi Sarcoma

Kaposi sarcoma is a low-to-intermediate-grade mesenchymal tumor and one of the defining illnesses of AIDS. Involvement of the GI tract is the most common manifestation of disseminated Kaposi sarcoma, and the duodenum is the most commonly affected site. Patients with Kaposi sarcoma may initially be asymptomatic, but hemorrhage, bowel obstruction, perforation, and malabsorption may occur as the lesions enlarge (57). Fluoroscopic findings are nonspecific and include ulceration, polypoid submucosal masses, and irregular fold thickening (Fig 29). The characteristic red-purple color of these lesions at endoscopy

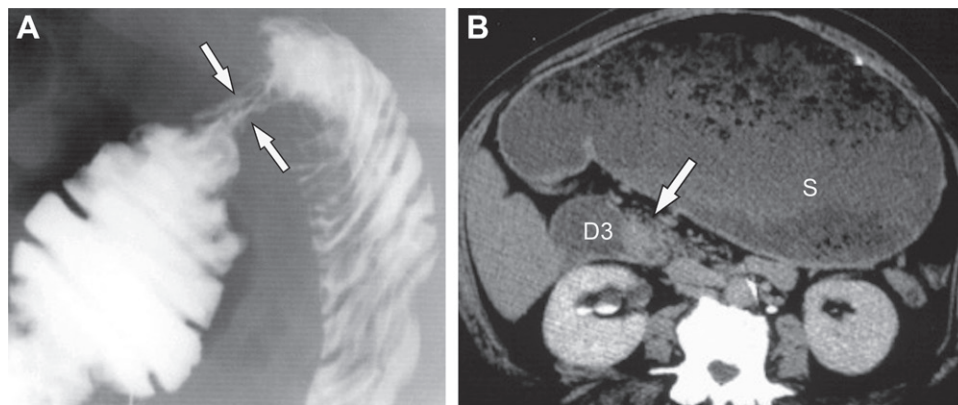


Figure 28. Metastasis from squamous cell carcinoma of the cervix in a 64-year-old woman with a history of bloating and cramping. **(A)** Fluoroscopic image shows focal narrowing of D3 (arrows). **(B)** Axial CT image (soft-tissue window) shows marked gastric distention and an intramural soft-tissue mass in D3 (arrow). S = stomach.

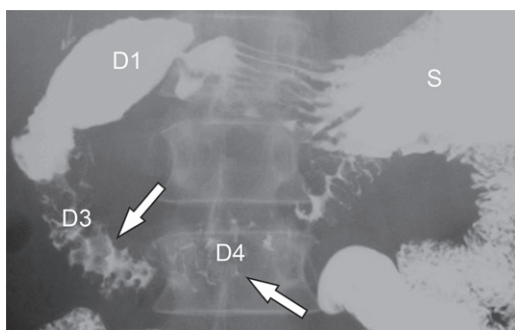


Figure 29. Kaposi sarcoma in a 42-year-old man with AIDS who presented with abdominal pain and diarrhea. Fluoroscopic image shows marked thickening of the mucosal folds, with polypoid filling defects involving D3 and D4 (arrows). S = stomach.

can be helpful in making the diagnosis, although biopsy results may be negative because of the submucosal location of the disease (58).

Other Tumors

Other uncommon tumors that can develop in the duodenum in rare cases include leiomyosarcoma, leiomyoma, plasmacytoma, and gangliocytic paraganglioma.

Miscellaneous Conditions

Superior Mesenteric Artery Syndrome

Superior mesenteric artery syndrome is characterized by compression of D3 between the superior mesenteric artery and the aorta, leading to narrowing and even obstruction. At CT, the diagnosis of superior mesenteric artery syndrome is made on the basis of an aortomesenteric angle of less than 22° and an aortomesenteric distance of less than 8 mm (59). Patients with substantial weight loss or a congenitally short ligament of Treitz and adolescents who have undergone corrective spinal surgery for scoliosis are at increased

risk of developing superior mesenteric artery syndrome (60). Presenting symptoms include early satiety, postprandial epigastric pain, nausea, and weight loss. Imaging findings indicative of superior mesenteric artery syndrome at fluoroscopy include dilatation of the stomach and proximal duodenum, with abrupt compression of D3 (Figs 30, E6). Active peristalsis may be observed as the dilated upstream segments of the duodenum attempt to propel enteric contents beyond the obstruction. Classically, bowel obstruction due to superior mesenteric artery syndrome is relieved when the patient is placed in the prone or left lateral decubitus position (59).

Hematoma

Anticoagulant medications, trauma, and bleeding disorders are the most common causes of a *duodenal hematoma*, a collection of blood in the duodenal wall (61). In infants and young children, a duodenal hematoma may raise suspicion for nonaccidental injury. At fluoroscopy, a duodenal hematoma may appear as segmental narrowing of the lumen with thickening of the folds or a focal area of scalloped luminal narrowing with fold effacement, most commonly affecting D3. Sufficiently large duodenal hematomas can result in obstruction of the gastric outlet or proximal small bowel (Figs 31, E7). The “stack of coins sign” describes smooth and uniform thickening of the small bowel folds at fluoroscopy and CT that can be seen in patients with small bowel hemorrhage.

Bouveret Syndrome

Bouveret syndrome is an uncommon variant of gallstone ileus, which occurs when a large gallstone erodes into the stomach or duodenum and causes obstruction of the outlet (Fig 32). It is much less common than classic gallstone ileus, where an obstructing stone is impacted in the

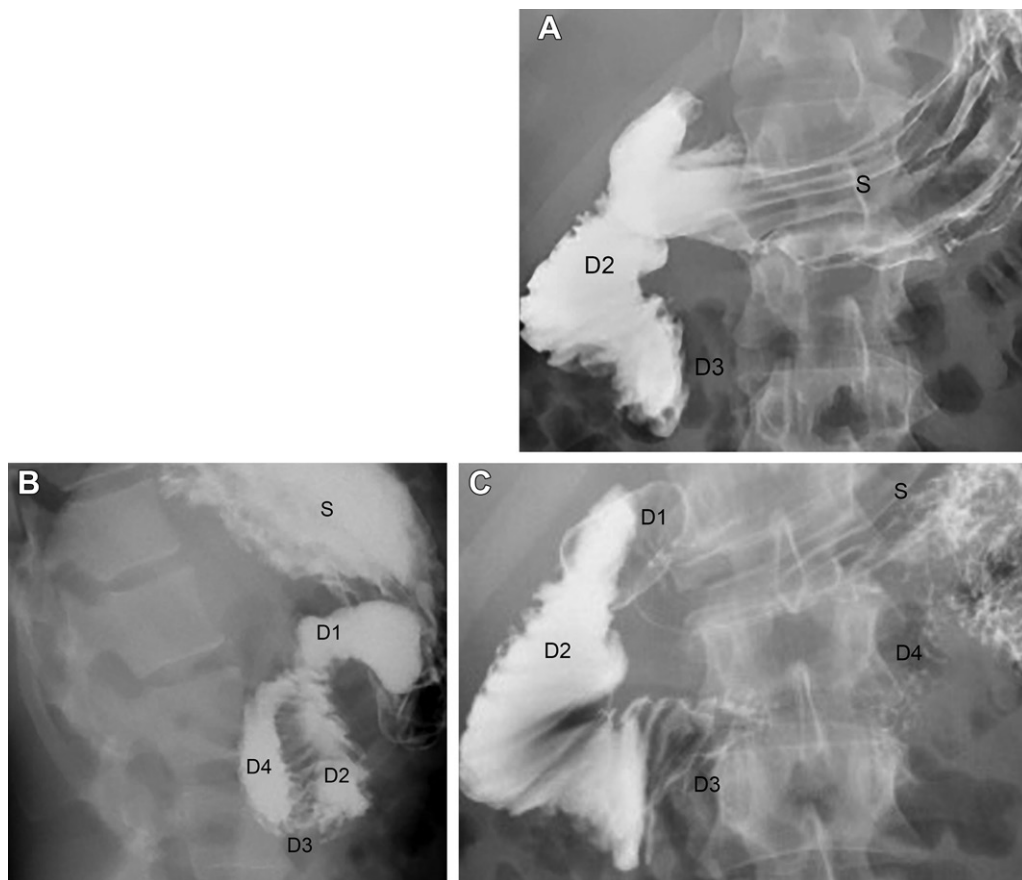


Figure 30. Superior mesenteric artery syndrome in a 22-year-old woman with abdominal pain. S = stomach. (A) Anteroposterior fluoroscopic image shows opacification of D1 and D2 and no flow of contrast material to D3, even on more delayed images. (B) Fluoroscopic image with the patient in the left lateral decubitus position shows that repositioning relieves the obstruction. (C) Follow-up fluoroscopic image with anteroposterior positioning shows opacification of the remainder of the duodenum.

terminal ileum (62). Bouveret syndrome results from chronic inflammation of the gallbladder and subsequent development of a cholecystoduodenal fistula. Presenting symptoms include nausea, emesis, and epigastric pain. In up to 35% of patients, the diagnosis may be suggested by the presence of the Rigler triad: bowel obstruction, pneumobilia, and an ectopic gallstone (63). During fluoroscopy, a large ovoid filling defect is found in the proximal duodenum, along with a fistulous tract between the duodenum and the gallbladder (64).

Varices

Duodenal varices are an uncommon cause of massive upper GI bleeding and are most common in D1. They result from portal hypertension, either related to hepatic cirrhosis or extrahepatic causes such as portal vein thrombosis (65). Norton et al (66), in a review of 169 patients, showed that up to 40% of patients with portal hypertension who undergo angiography have duodenal varices. At fluoroscopy, varices manifest as serpentine or tortuous filling defects

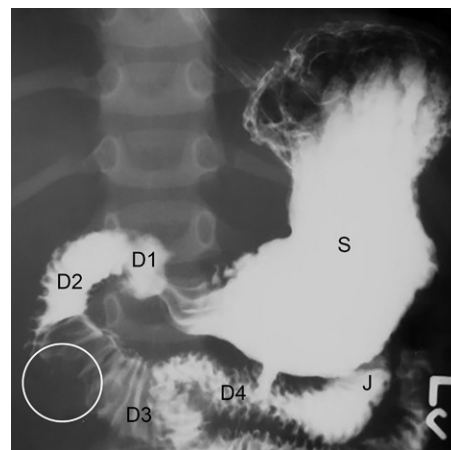


Figure 31. Duodenal mural hematoma in a 6-year-old boy after blunt trauma. Fluoroscopic image shows narrowing of the duodenal lumen at the junction of D2 and D3 from a circumscribed mural mass along the lateral wall that forms acute margins with the normal lumen (circle). The linear collection of barium that appears to arise from the greater curvature of the stomach (S) likely represents superimposition of a fold near the ligament of Treitz. There is no extraluminal contrast material. J = jejunum.

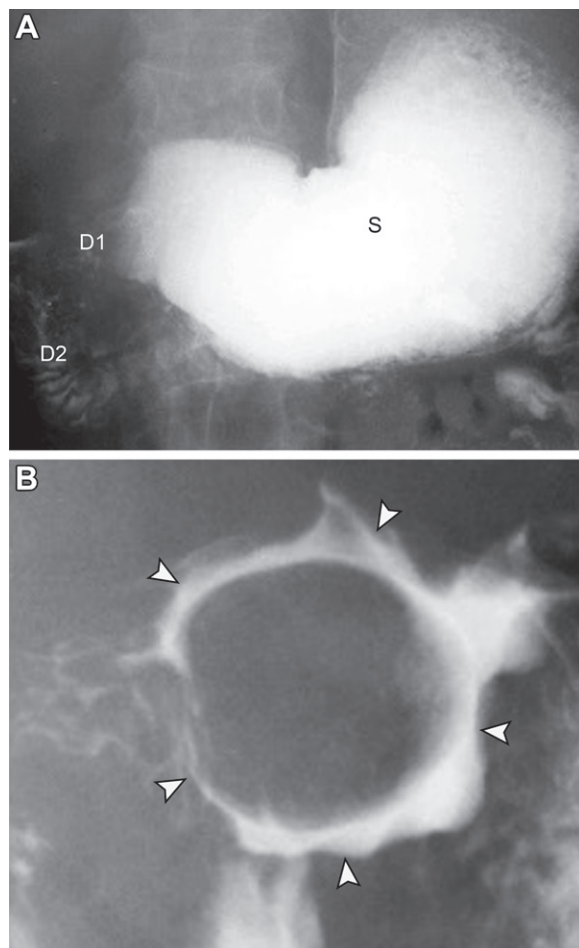


Figure 32. Bouveret syndrome in an 83-year-old woman with 1 week of upper abdominal pain, nausea, and vomiting. (A) Anteroposterior fluoroscopic image of the upper GI tract shows distended fluid and debris filling the stomach (S). (B) Coned-down fluoroscopic image shows a 5-cm rounded lucent filling defect (arrowheads) in the distorted D1.

that may disappear with luminal distention. Patients with hereditary hemorrhagic telangiectasia can also present with hematemesis (or recurrent pancreatitis) because of vascular anomalies, specifically arteriovenous malformations.

Postoperative Leaks

Like any hollow viscus organ, the duodenum is susceptible to perforation. This can occur because of blunt or penetrating trauma or as a complication of abdominal surgery, resulting in leakage of enteric contents into the peritoneum. The Billroth II gastrojejunostomy is a procedure performed for disease in the distal stomach and involves partial gastrectomy and a side-to-side anastomosis with the residual stomach and the first part of the jejunum. The procedure creates two limbs of the small bowel, an afferent and an efferent limb, with the afferent limb (also known as the upstream or biliopancreatic limb) primarily comprising the duodenal stump, which transports



Figure 33. Duodenal stump leak in a 65-year-old man after he underwent the Billroth II procedure for gastric ulcers and presented with abdominal pain and sepsis. Fluoroscopic image shows leakage of oral contrast material (arrow) from the duodenal stump. D = duodenum, S = stomach.

pancreaticobiliary secretions toward the stomach. A known complication of the Billroth II procedure is a duodenal stump leak, which occurs in 1%–3% of patients and has high morbidity and mortality rates (67). Extraluminal contrast material leakage during fluoroscopy is diagnostic of this complication, although a study that is negative for contrast extravasation does not exclude the presence of a small leak, because opacification of this limb requires retrograde flow (Fig 33). If clinical suspicion is high and the fluoroscopy results are negative, further evaluation may be warranted with CT or endoscopy. Duodenal leaks may also occur in patients who have undergone reconstructive GI surgery, because anastomotic leaks may result in an abscess or fluid collection in the surgical bed (68).

Afferent Loop Syndrome

Afferent loop syndrome is obstruction and dilatation of the afferent loop, which occurs in up to 2% of patients after they undergo gastroenterostomy. Patients present clinically with abdominal pain, nausea, and vomiting (69). Other surgical procedures that can lead to afferent loop syndrome include Roux-en-Y esophagojejunostomy, Roux-en-Y gastric bypass, and pancreatic surgical procedures such as the Whipple procedure (70). Three mechanisms that contribute to afferent loop syndrome include obstruction of the afferent loop; obstruction of the efferent loop, leading to preferential filling of the afferent loop; and preferential gastric emptying into the afferent loop. Isolated obstruction of the afferent loop is a



Figure 34. Afferent loop syndrome in a 72-year-old man after he underwent the Billroth II procedure for gastric carcinoma and presented with increasing nausea, vomiting, and weight loss. Anteroposterior fluoroscopic image of the upper GI tract shows marked dilatation of the duodenum (afferent limb) (arrowheads) with no contrast material or gas in the distal small bowel.

closed loop obstruction requiring urgent surgical evaluation. At fluoroscopy, contrast material fails to enter the afferent limb, although the normal nonobstructed afferent limb may not fill in 20% of patients. Alternatively, preferential filling and retention of enteric contrast material in the afferent limb for at least 60 minutes is consistent with afferent loop syndrome, which may be due to efferent loop obstruction (Fig 34) (71).

Other

A characteristically abnormal appearance of the duodenum at fluoroscopy is widening of the duodenal sweep, usually because of a pancreatic abnormality such as pseudocysts, neoplasms, and inflammation from pancreatitis (Fig 35) (61). Other miscellaneous abnormalities that affect the duodenum include celiac sprue disease, eosinophilic duodenitis, aortoenteric fistula, and immunoglobulin G4-related disease.

Conclusion

Many duodenal abnormalities can be diagnosed with fluoroscopy. In other cases, fluoroscopy may suggest a group of diseases such as malignancy or widespread inflammation and help to guide patient care. Although the fluoroscopic imaging features of many duodenal abnormalities may overlap, clinical context and subsequent cross-sectional imaging or evaluation with endoscopy can further narrow the differential diagnosis. Fluoroscopy is especially helpful in the postoperative abdomen because it allows delineation of the altered anatomy and identification of common postoperative complications. The anatomic and functional information elicited from fluoroscopy is a valuable adjunct to endoscopy and cross-sectional imaging. Radiologists' familiarity with fluoroscopic techniques, normal anatomy, and imaging features of common duo-

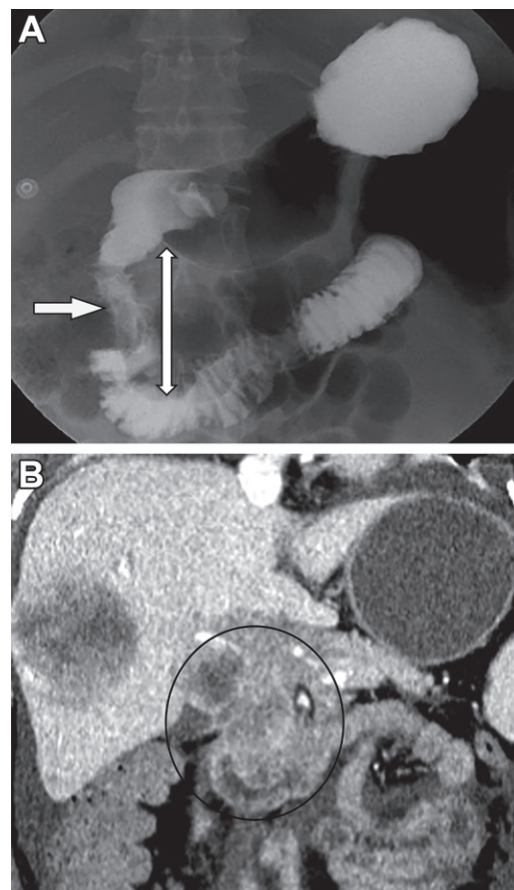


Figure 35. Widening of duodenal sweep in a 40-year-old woman with a metastatic pancreatic neuroendocrine tumor. (A) Anteroposterior fluoroscopic image shows widening of the duodenal sweep (double-headed arrow) and persistent narrowing of D2 (arrow). (B) Axial CT image (soft-tissue window) shows an ill-defined mass in the head of the pancreas (circle) and hepatic metastasis (arrowhead).

denal abnormalities is important in making timely diagnoses and guiding appropriate management.

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